

Stat 150 – Lecture 02

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1 Markov Chain 2.0

1.1 Classification of States:

Understand the behavior of Markov chains. Sometimes they are recurrent, sometimes transient, etc.

Notation: $i \rightarrow j$ denotes “ j is reachable or accessible from i ”, that is

$\exists n \in \mathbb{Z}^+$ such that $P_{ij}^n =$ probability of jump to j at n steps starting from $i > 0$

Write $i \leftrightarrow j$ if $i \rightarrow j$ and $j \rightarrow i$ (they **communicate**). Communication is an equivalence relation, so it splits the state space into **communication classes**; the chain is **irreducible** if there is only one class, the whole state space.

Definition 1.1 (Chapman–Kolmogorov Equation – Discrete Case). To travel from i to k in $m + n$ steps, the chain passes through some intermediate state r at time m ; summing over all intermediates,

$$P_{ik}^{m+n} = \sum_r P_{ir}^m P_{rk}^n$$

In matrix form this is exactly $P^{m+n} = P^m P^n$: multi-step transition matrices compose by ordinary matrix multiplication.

Notation:

1. $\mathbb{P}_x(A) = \mathbb{P}(A \mid x_0 = x)$ for an event A in a markov chain $(x_n)_{n \in \mathbb{N}}$
2. Denote $T_y = \min\{n \geq 1 : X_n = y\}$ time of the first return to y . (y at time 0, return back).
- 3.

$$\rho_{yy} = \mathbb{P}_y(T_y < \infty) = \sum_{n=1}^{\infty} \mathbb{P}_y(T_y = n)$$

is the probability that, starting at y , we return in a finite number of steps. (Durrett writes this ρ_{yy} , “rho”, rather than P_{yy} , to avoid confusion with the transition probabilities $P(i, j)$.)

Definition 1.2 (Recurrent and Transient States). A state y is **recurrent** if $\rho_{yy} = 1$ and **transient** if $\rho_{yy} < 1$.

Why this dichotomy is the whole story: by the strong Markov property, each return to y “restarts” the chain, so successive returns behave like independent trials with success probability ρ_{yy} . The probability of a k th return is $(\rho_{yy})^k$, the probability of exactly k returns is $(\rho_{yy})^k(1 - \rho_{yy})$, and the expected number of returns is $\rho_{yy}/(1 - \rho_{yy})$. The chain returns to y infinitely often with probability 1 *exactly when* $\rho_{yy} = 1$ – there is no middle ground. (This is Durrett §1.3, and it is where the “ k th return is $(\rho_{yy})^k$ ” claim from lecture comes from.)

Definition 1.3 (Strong Markov Property). The memoryless property survives stopping at *random* times: stopped at a stopping time and restarted, the chain is again a Markov chain with the same transitions, independent of the past. This is what powers results like “the probability of a k th return is $(\rho_{yy})^k$.”

Definition 1.4 (Stopping Time). A random time T (often written τ) is a stopping time if for every n the event $\{\tau = n\}$ (“we stop at time n ”) is determined by $X_0, X_1, \dots, X_n$ only – no peeking at the future. Examples: the first return time T_y ; the first time the chain enters a set A of states.

1.2 Durrett: Strong Markov Chain Properties Let $(X_t)_{t \in \mathbb{N}}$ be a Markov chain, τ a stopping time, and E an event determined by the history up to τ . Then for all $i, j \in S$ and $k \geq 1$,

$$\mathbb{P}(X_{\tau+k} = j \mid X_\tau = i, E) = \mathbb{P}(X_{\tau+k} = j \mid X_\tau = i),$$

and the shifted chain $(Y_n) := X_{\tau+n}$, $n \geq 0$, is itself a Markov chain with the same transition matrix.

Sketch of Proof: condition on $\{\tau = n\}$ (allowed, because $\{\tau = n\}$ is determined by $X_0, \dots, X_n$) and apply the ordinary Markov property. We eventually show

$$\mathbb{P}(X_{\tau+1} = z \mid X_\tau = y, \tau = n) = P_{yz} = P(y, z),$$

then sum over n to drop the conditioning on the particular stopping time.

2 Questions and follow-up

- Connect the lectures: for the symmetric walk on $\mathbb{Z}$ (recurrent), what is ρ_{yy} , and why can the expected return time still be infinite?
- Redo Durrett's §1.3 computation showing transience when $p \neq \frac{1}{2}$.
- TODO: